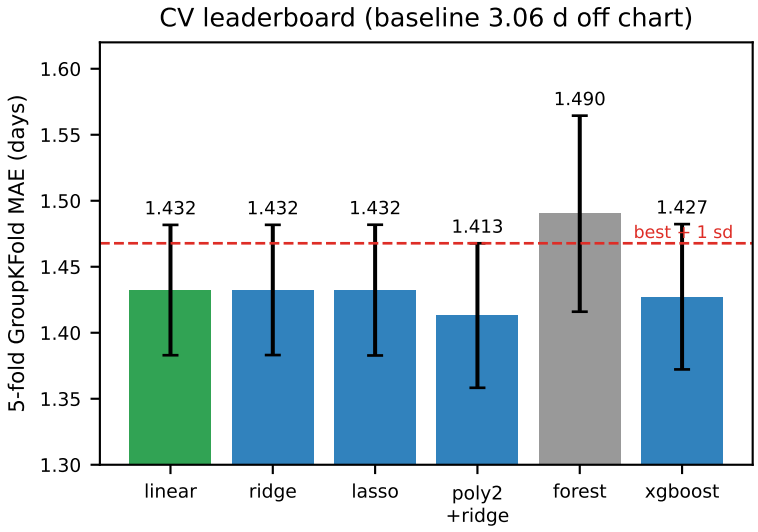
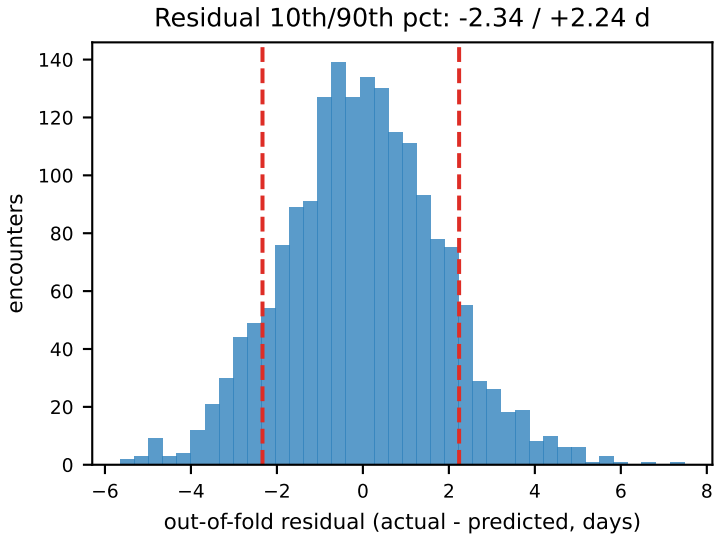
Day 28 Cheat Sheet: The Week 4 Regression Scorecard

Week 4 wrap-up: one fair race, a rule written first, the test set opened once, a model card

STEP	WHAT TO DO	WHY
1 Same everything	same features, patient split, GroupKFold(5) folds	differences come from models, not luck
2 Pipelines	impute + scale + model inside one Pipeline	no preprocessing leak (day 27)
3 Many metrics	cross_validate: MAE, RMSE, R2, fit time	MAE for planners, RMSE punishes misses
4 Baseline first	DummyRegressor(median) sets the floor	"better than guessing" by how much?
5 Rule before test	best CV MAE, then simplest within 1 sd	avoids picking noise; easier to explain
6 Test once	refit on all TRAIN, score test, never tune after	an honest final number
7 Uncertainty	80% band from out-of-fold residual quantiles	a range planners can use
8 Subgroups	error by acuity (or age, unit, payer)	find where the model is weakest
9 Save + document	joblib.dump + model_card.json	reproducible and reviewable





THIS DATA: 2,400 simulated encounters from 800 patients (3 stays each, seed 28), 8 admission-time features, albumin 12% missing, one built-in ICU x procedure interaction. Split by patient: train 1,800, test 600. Mean LOS 5.79 days.
CV MAE (days): baseline 3.065 | linear 1.432 | ridge 1.432 | lasso 1.432 | poly2+ridge 1.413 | forest 1.490 | xgboost 1.427
Best: poly2 + ridge (1.413 +/- 0.055). Within 1 sd: 5 models. Champion by the rule: linear.
Test once: MAE 1.534 days, RMSE 1.941, R2 0.723 (baseline MAE 3.083). 80% band covered 77.8% of test stays.
Test came out worse than CV (1.534 vs 1.432): one split of 200 test patients is noisy, so report both and quote CV as the estimate.

RULES OF THUMB

- A gap smaller than the fold-to-fold standard deviation is a lean, not a win. Prefer the model you can explain.
- Report MAE in the unit people use (days) next to the baseline, so "1.4 days" has context.
- A point estimate alone hides risk. Add a band and check its coverage on the test set.
- The model card is part of the model: data, metric, intended use, and what it must not be used for.

```
cv = cross_validate(pipe, X, y, groups=patient_id, cv=GroupKFold(5),
        scoring={"mae": "neg_mean_absolute_error", "r2": "r2"})
oof = cross_val_predict(pipe, X, y, groups=patient_id, cv=GroupKFold(5))
lo, hi = np.quantile(y - oof, [0.10, 0.90])   # 80% band
```

Infographic: Picking a length-of-stay model you can defend

Seven models, one fair race, one honest test

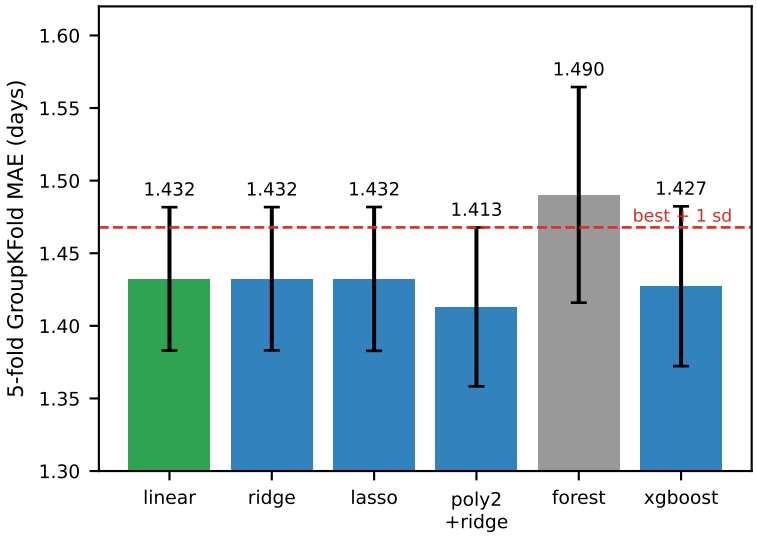
BASELINE MAE
3.06 d

BEST CV MAE
1.41 d

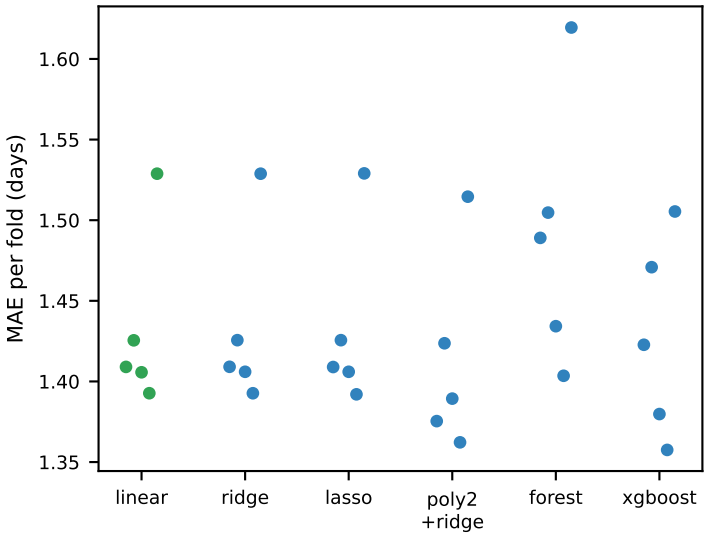
CHAMPION
LINEAR

TEST MAE
1.53 d

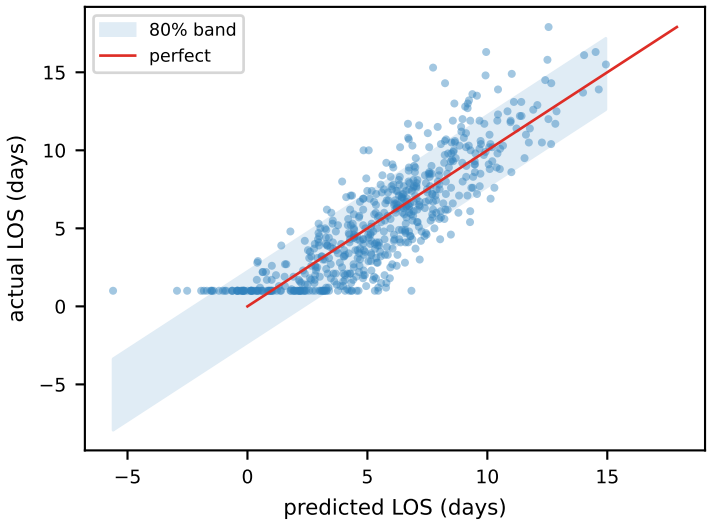
CV leaderboard (baseline 3.06 d off chart)



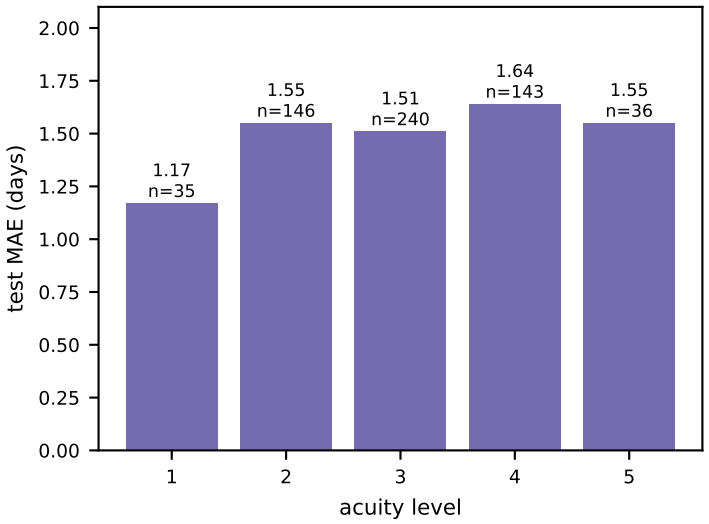
Fold to fold spread is bigger than the gaps



Test set, once: MAE 1.53, R2 0.72, band covers 78%



Error by subgroup




```
MODEL CARD  (model_card.json)
Model:      linear regression (Pipeline: impute, scale, fit)
Target:     length of stay in days, predicted at admission
Data:       2,400 SIMULATED encounters, 800 patients, split by patient
Choice:     simplest model within 1 sd of best CV MAE
CV MAE:     1.432 +/- 0.049 days
Test:       MAE 1.534, RMSE 1.941, R2 0.723
Band:       80% interval pred -2.34/+2.24 d, covers 77.8%
Use:        teaching, bed-planning style estimates
Not for:    individual clinical decisions
Limits:     simulated, no time split, frailty unseen
```

How to read it

1. Five models sit within one standard deviation of the best (poly2 + ridge, 1.413 d).
2. The rule picks the simplest: linear. Fewer moving parts, readable coefficients.
3. Test MAE 1.53 is worse than CV 1.43: one split varies.
4. Every model halves the baseline error (3.06 d).

Hands-On Worksheet: ds-genai-90-days / day28

60-90 minutes | notebook: day28/los_scorecard.ipynb + day28/README.md + model_card.json

[ ] mkdir day28, conda activate ds90, open day28/los_scorecard.ipynb

[ ] Simulate 2,400 encounters from 800 patients, seed 28 (snippet 1)

[ ] Split by patient with GroupShuffleSplit; set up GroupKFold(5)

[ ] Build 7 Pipelines, simplest to most complex (snippet 2)

[ ] cross_validate each: MAE, RMSE, R2, fit time; print the leaderboard

[ ] Write your selection rule in a markdown cell BEFORE going further

[ ] Apply it: best CV MAE, then simplest within 1 sd (snippet 3)

[ ] Plot per-fold MAE for every model; is the gap bigger than the spread?

[ ] Out-of-fold residuals on TRAIN -> 10th/90th percentile band (snippet 4)

[ ] Refit the champion on TRAIN, score the test set ONCE

[ ] Check band coverage and error by acuity level

[ ] joblib.dump the model; write model_card.json (snippet 5)

[ ] README: leaderboard, rule, test numbers, card, limits (simulated)

[ ] git add day28 && git commit && git push

[ ] Stretch: store CV results in a Snowflake table (snippet 7)

```
best = board.cv_mae.idxmin()
limit = board.loc[best, "cv_mae"] + board.loc[best, "mae_sd"]
eligible = [m for m in cands if board.loc[m, "cv_mae"] <= limit]
champion = eligible[0]            # dict is ordered simplest -> most complex
```

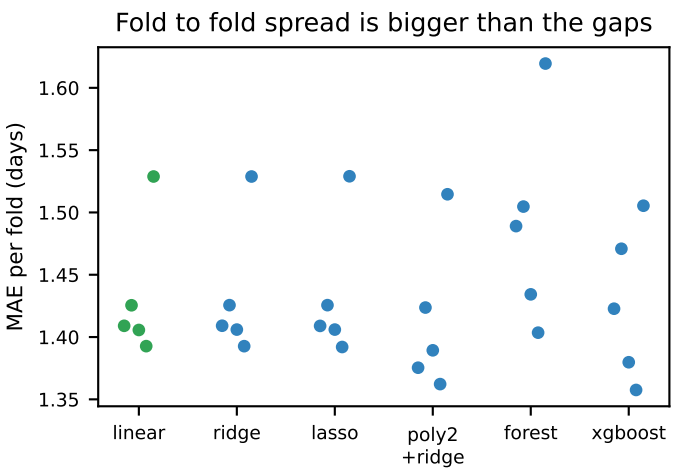
Baseline CV MAE: _____ Best model: _____ _____

Within 1 sd: _____

Champion: _____ Test MAE: _____ R2: _____

80% band: ____ / ____ coverage: _____

Notes: _____



Day 28 Self-Check Quiz

1. Why must every model in the race use the same folds and the same patient split?

_____

2. Two models differ by 0.02 days of CV MAE with a fold sd of 0.05. What do you conclude?

_____

3. Why write the selection rule before opening the test set?

_____

4. Test MAE is worse than CV MAE. Did something break? What do you report?

_____

5. What does a DummyRegressor baseline add to the scorecard?

_____

6. How do you build an 80% prediction band without touching the test set?

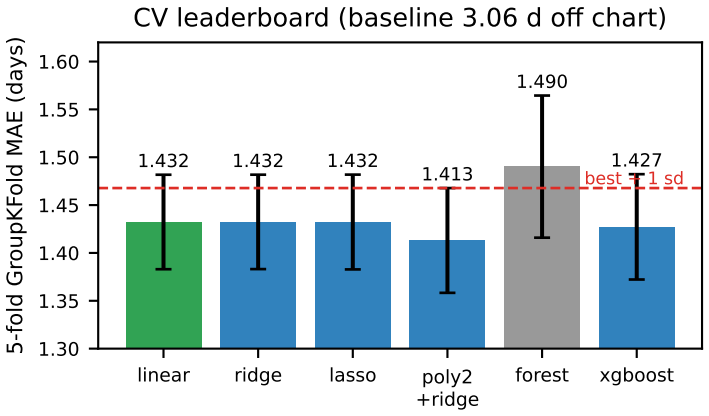
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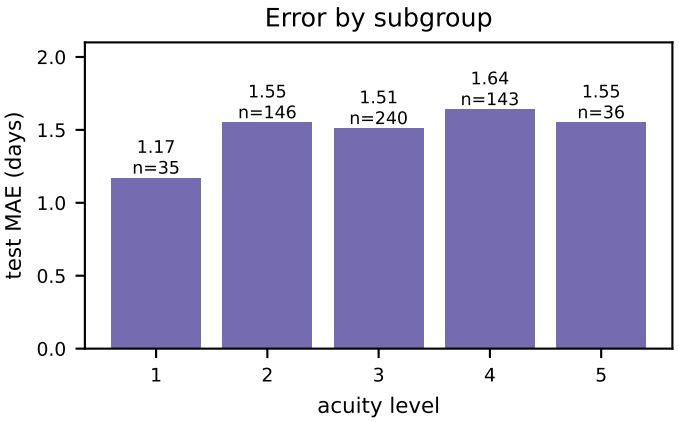
7. Name four things a model card should contain.

_____

8. Why check error by acuity (or another subgroup) before shipping?

_____





Answers:
1 so differences come from the models, not from easier folds; GroupKFold keeps each patient in one fold
2 it is a lean, not a win: the gap is inside the noise, so prefer the simpler, more explainable model
3 if you choose after seeing test scores, the test set becomes another validation set and the number is optimistic
4 not necessarily: one 200-patient split is noisy (1.534 vs CV 1.432); report both, quote CV, never retune
5 the floor: guessing the median gives 3.06 days MAE, so the model's 1.43 has context
6 take out-of-fold residuals on TRAIN, use their 10th and 90th percentiles around each prediction (-2.34/+2.24)
7 any four: data and split, features, metrics with CV and test, intended use, out-of-scope uses, limits, versions
8 an average can hide a weak group; here acuity 4 had the highest MAE (1.64 days)